

Fourth Semester B. Tech. (Electronics and Communication Engineering / Electronics Design Technology) Examination**INTRODUCTION TO ELECTROMAGNETIC THEORY**

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) Assume suitable data wherever necessary. Consider bold letters as a vector.
- (3) Given Constants : $\epsilon_0 = 8.854 \times 10^{-12}$ F/m and $\mu_0 = 4\pi \times 10^{-7}$ H/m.

1. Solve any Six questions :

- (A) Transform vector $10\mathbf{a}_x$ into cylindrical coordinate system at point $Q(x = 3, y = 4, z = 1)$ 2(CO1)
- (B) Write divergence operator for orthogonal curvilinear coordinate system and give it in cylindrical form with its significance. 2(CO1)
- (C) The electric field intensity due to infinite uniform sheet charge is given as $\mathbf{E}_s = (\rho_s / 2\epsilon) \mathbf{a}_n$. Justify this statement with suitable diagram. 2(CO1)
- (D) How Poynting theorem explains flow of electricity through conductor ? 2(CO2)
- (E) Give properties of good conductor. 2(CO4)
- (F) What is uniform plane wave ? 2(CO4)
- (G) Give the Magnetic boundary conditions at the interface of two different magnetic materials. 2(CO3)
- (H) If $\phi_B = 5t^2 + 4t + 1$. Find out the induced emf through the coil at $t = 2$ seconds. 2(CO3)

2. (A) Find the charge density using Gauss law for $\mathbf{D} = 2\rho \cos\phi \mathbf{a}_\phi - (\sin\phi/3\rho) \mathbf{a}_z$ Over the surface placed at $Z = 0$ plane. ρ value is between 0 to a , and $\phi = 0$ to $\pi/2$. 3(CO2)

- (B) Given that $\mathbf{D} = 20\rho^2 \mathbf{a}_\rho$ C/m². The surface is bounded with $0 \leq \rho \leq 3$ and $0 \leq z \leq 2$. Evaluate either side of the Gauss Divergence theorem. 4(CO2)
- (C) Deduce the expression of the potential difference between two points A and B and hence give the importance of concept zero potential. 3(CO3)
3. (A) Given a non uniform field $\mathbf{E} = y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z$. Determine the work expended in carrying a charge of 2C from B (1, 0, 1) to A (0.8, 0.6, 1) along the shorter arc of the circle $x^2 + y^2 = 1$, $Z = 1$. 4(CO2,3)
- (B) Let an infinite current element drawing a current $|\mathbf{d}| = 6\mathbf{a}_z$ A-m. Find out the magnetic field intensity $\mathbf{H}$ at point P(2, 3, 4). 3(CO2)
- (C) Obtain an expression for Electric field due to atomic dipole. Further find the field on the axis of the dipole. 3(CO2,3)
4. (A) Calculate the value of the vector current density :
 (i) In cartesian coordinate at $P_A(3, 2, 1)$ if vector $\mathbf{G} = xyz(\mathbf{a}_x + \mathbf{a}_y)$.
 (ii) In cylindrical coordinate at $P_B(2, 30^\circ, 3)$ if $\mathbf{G} = \rho\phi z \cdot \mathbf{a}_z$. 4(CO1,2)
- (B) The magnetic field intensity is given by, $\mathbf{H} = (2\rho - \rho^2)\mathbf{a}_\phi$ A/m for $0 < \rho < 1$. Determine :
 (i) The current density as a function of ρ with cylinder.
 (ii) What is the total current passing through the surface $z = 0$, $0 < \rho < 1$ in $\mathbf{a}_z$ direction ? 6(CO2,4)
5. (A) Obtain vector Helmholtz equation from phasor equation for electric field. 4(CO4)
- (B) What is scalar magnetic potential ? Give its significance and comment on its multiple values using suitable example. 3(CO3)
- (C) Discuss uniform plane wave propagating in good conducting medium. Hence explain in brief the meaning of skin effect. 3(CO4)

6. Write short notes on any **Two** :—

- (i) Isotropic and homogeneous medium.
- (ii) Vector magnetic potential.
- (iii) Poynting theorem.
- (iv) Electric field at the boundary when it is incident normally on a plane conducting boundary.

8(CO3,4)

